

Irman Sazzad

The planner chooses c_t , l_t , k_t to maximize

$$\sum_{t=0}^{\infty} \beta^t [\gamma \ln c_t + (1 - \gamma) \ln(1 - l_t)]$$

subject to the period budget constraint

$$y_t = c_t + i_t, \quad y_t = A_t k_{t-1}^\alpha l_t^{1-\alpha},$$

and the capital accumulation equation

$$k_t = (1 - \delta)k_{t-1} + i_t.$$

Substituting $i_t = k_t - (1 - \delta)k_{t-1}$ into the resource constraint gives

$$c_t + k_t = A_t k_{t-1}^\alpha l_t^{1-\alpha} + (1 - \delta)k_{t-1}.$$

The Lagrangian for the planner is

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t [\gamma \ln c_t + (1 - \gamma) \ln(1 - l_t) + \lambda_t (A_t k_{t-1}^\alpha l_t^{1-\alpha} + (1 - \delta)k_{t-1} - c_t - k_t)].$$

The Lagrangian from part (i) is

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t [\gamma \ln c_t + (1 - \gamma) \ln(1 - l_t) + \lambda_t (A_t k_{t-1}^\alpha l_t^{1-\alpha} + (1 - \delta)k_{t-1} - c_t - k_t)].$$

The first order condition with respect to consumption is

$$\frac{\partial \mathcal{L}}{\partial c_t} = \beta^t \left(\frac{\gamma}{c_t} - \lambda_t \right) = 0 \quad \Rightarrow \quad \lambda_t = \frac{\gamma}{c_t}.$$

Here γ/c_t is the marginal cost of postponing one unit of consumption in period t , because the parameter γ determines the weight placed on consumption in period utility and the term $1/c_t$ captures the curvature of utility, making the cost larger when consumption is low. The multiplier λ_t is the marginal benefit of saving that unit, since it measures the shadow value of additional resources in the constraint and therefore indicates how valuable one more unit of output is for future production through its contribution to the capital stock. Equality between γ/c_t and λ_t ensures that the utility cost of delaying consumption exactly matches the value created by increasing future productive capacity.

The first order condition with respect to labor is

$$\frac{\partial \mathcal{L}}{\partial l_t} = \beta^t \left(\frac{-(1 - \gamma)}{1 - l_t} + \lambda_t (1 - \alpha) A_t k_{t-1}^\alpha l_t^{-\alpha} \right) = 0.$$

Here $\frac{-(1-\gamma)}{1-l_t}$ is the marginal loss in period utility from supplying an additional unit of labor, with the parameter $(1 - \gamma)$ determining the weight on leisure and the term $1/(1 - l_t)$ capturing the sensitivity of utility to reductions in leisure. The expression $\lambda_t (1 - \alpha) A_t k_{t-1}^\alpha l_t^{-\alpha}$ is the marginal benefit of additional labor, since $(1 - \alpha)$ reflects labor's output share in the production function, A_t measures the productivity of the economy, and the multiplier λ_t converts the resulting gain in output into its value in the resource constraint. The condition requires the utility cost of reducing leisure to equal the value of the extra output generated by increasing labor input.

The first order condition with respect to capital is

$$\frac{\partial \mathcal{L}}{\partial k_t} = -\beta^t \lambda_t + \beta^{t+1} \lambda_{t+1} [\alpha A_{t+1} k_t^{\alpha-1} l_{t+1}^{1-\alpha} + 1 - \delta] = 0,$$

which implies

$$\lambda_t = \beta \mathbb{E}_t [\lambda_{t+1} (\alpha A_{t+1} k_t^{\alpha-1} l_{t+1}^{1-\alpha} + 1 - \delta)].$$

Here λ_t is the marginal benefit of saving one unit of output in period t , because it measures the value of relaxing the resource constraint and therefore the value of adding to the capital stock. The right hand side represents the discounted expected marginal benefit of this saved unit in period $t + 1$, with the gross return on capital given by $\alpha A_{t+1} k_t^{\alpha-1} l_{t+1}^{1-\alpha} + 1 - \delta$, where α is the capital share in production, A_{t+1} is next period productivity, and $1 - \delta$ captures the portion of capital that remains after depreciation. Multiplying this return by λ_{t+1} expresses the value of the additional resources generated in the next period, and the factor β discounts this benefit to period t . The condition equates the current marginal benefit of saving with the expected discounted future marginal benefit generated through capital accumulation.

Substituting $\lambda_t = \gamma/c_t$ into the capital condition yields the Euler equation,

$$\frac{\gamma}{c_t} = \beta \mathbb{E}_t \left[\frac{\gamma}{c_{t+1}} (\alpha A_{t+1} k_t^{\alpha-1} l_{t+1}^{1-\alpha} + 1 - \delta) \right].$$

Here, γ/c_t is the marginal cost of postponing one unit of consumption, which is larger when consumption is low or when the utility weight on consumption γ is high. The right hand side is the expected present value of the marginal benefit of saving that unit, where the return on capital depends on the capital share α , the future productivity level A_{t+1} , the labor input l_{t+1} , and the depreciation rate δ , and where the shadow value γ/c_{t+1} indicates how valuable the additional output is in the next period. The discount factor β adjusts this benefit for time preference. The Euler equation therefore requires that the cost of postponing consumption today equals the discounted expected return generated by increasing next period capital.

The equilibrium conditions derived in parts (i) and (ii) allow for a numerical representation of the social planner's problem in Dynare. The calibration follows the given parameter values and maintains internal consistency with the analytical structure of the model. The discount factor is set to $\beta = 0.97$, the capital share to $\alpha = 0.35$, the depreciation rate to $\delta = 0.06$, and the utility weight on consumption to $\gamma = 0.40$. The exogenous productivity variable A_t evolves according to the autoregressive process

$$\ln A_t = (1 - \rho_A) \ln \bar{A} + \rho_A \ln A_{t-1} + \varepsilon_t,$$

with steady-state productivity $\bar{A} = 1$, persistence parameter $\rho_A = 0.95$, and innovation term ε_t representing the technological shock, with variance σ_A^2 and standard deviation $\sigma_A = 0.01$. The logarithm of productivity is denoted by $z_t = \ln A_t$, which allows the productivity process to be written equivalently as

$$z_t = \rho_A z_{t-1} + \varepsilon_t,$$

and productivity is recovered from the relation $A_t = e^{z_t}$.

The Dynare implementation reproduces the optimality conditions implied by the planner's problem: the production function, the resource constraint, the capital accumulation equation, the marginal utility of consumption, the labour supply condition, and the Euler equation. The system is solved around the nonstochastic steady state and then the model is linearised through first-order perturbation, and the impulse responses reported in part (iv) are generated from this approximation.

```
% -----
% Real Business Cycle Model
% -----

% 1. Parameters
parameters beta gamma alpha delta rho_A sigma_A;
beta      = 0.97;
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```

gamma    = 0.40;
alpha    = 0.35;
delta    = 0.06;
rho_A    = 0.95;
sigma_A  = 0.01;

% 2. Endogenous variables
var c l k y i A z lambda;

% 3. Exogenous shock
varexo eps_A;

% -----
% 4. Model equations
% -----

model;

    % TFP (log form)
    z = rho_A*z(-1) + eps_A;
    A = exp(z);

    % Production function
    y = A * k(-1)^alpha * l^(1-alpha);

    % Resource constraint
    y = c + i;

    % Capital accumulation
    k = (1-delta)*k(-1) + i;

    % FOC: consumption

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lambda = gamma/c;

% FOC: labor
(1-gamma)/(1-l) = lambda*(1-alpha)*y/l;

% Euler equation
lambda = beta*lambda(+1)*( alpha*y(+1)/k + 1-delta );
end;

% -----
% 5. Initial values
% -----
initval;
    z      = 0;
    A      = 1;
    l      = 0.36;
    k      = 2.8;
    y      = 0.75;
    c      = 0.57;
    i      = 0.18;
    lambda = 0.7;
end;

steady;
check;

% -----
% 6. Shock variance
% -----
shocks;

```

```

    var eps_A = sigma_A^2;
end;

% -----
% 7. Simulation
% -----
stoch_simul(order=1, irf=40);

```

STEADY-STATE RESULTS:

c	0.572708
l	0.360396
k	2.86649
y	0.744697
i	0.171989
A	1
z	0
lambda	0.698437

EIGENVALUES:

Modulus	Real	Imaginary
0.8866	0.8866	0
0.95	0.95	0
1.163	1.163	0
6.014e+16	6.014e+16	0

There are 2 eigenvalue(s) larger than 1 in modulus for 2 forward-looking variable(s)
The rank condition is verified.

MODEL SUMMARY

Number of variables: 8
 Number of stochastic shocks: 1
 Number of state variables: 2
 Number of jumpers: 2
 Number of static variables: 4

MATRIX OF COVARIANCE OF EXOGENOUS SHOCKS

Variables eps_A
 eps_A 0.000100

POLICY AND TRANSITION FUNCTIONS

	c	l	k	y	i	A	z	lambda
Constant	0.572708	0.360396	2.866490	0.744697	0.171989	1.000000	0	0.698437
k(-1)	0.108550	-0.026607	0.886642	0.055192	-0.053358	0	0	-0.132380
z(-1)	0.310584	0.160849	0.612917	0.923501	0.612917	0.950000	0.950000	-0.378768
eps_A	0.326931	0.169315	0.645176	0.972106	0.645176	1.000000	1.000000	-0.398703

THEORETICAL MOMENTS

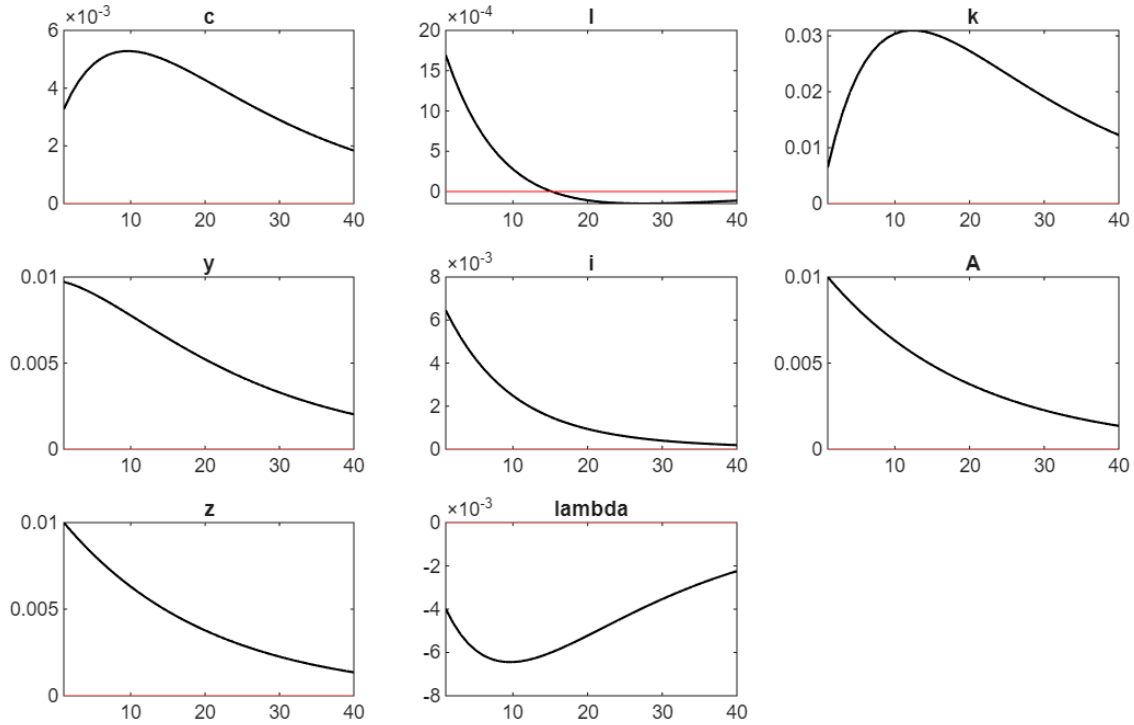
VARIABLE	MEAN	STD. DEV.	VARIANCE
c	0.5727	0.0258	0.0007
l	0.3604	0.0032	0.0000
k	2.8665	0.1527	0.0233
y	0.7447	0.0383	0.0015
i	0.1720	0.0148	0.0002
A	1.0000	0.0320	0.0010
z	0.0000	0.0320	0.0010
lambda	0.6984	0.0315	0.0010

MATRIX OF CORRELATIONS

Variables	c	l	k	y	i	A	z	lambda
c	1.0000	0.3429	0.9875	0.9681	0.7612	0.9283	0.9283	-1.0000
l	0.3429	1.0000	0.1903	0.5672	0.8702	0.6676	0.6676	-0.3429
k	0.9875	0.1903	1.0000	0.9165	0.6492	0.8580	0.8580	-0.9875
y	0.9681	0.5672	0.9165	1.0000	0.8993	0.9918	0.9918	-0.9681
i	0.7612	0.8702	0.6492	0.8993	1.0000	0.9477	0.9477	-0.7612
A	0.9283	0.6676	0.8580	0.9918	0.9477	1.0000	1.0000	-0.9283
z	0.9283	0.6676	0.8580	0.9918	0.9477	1.0000	1.0000	-0.9283
lambda	-1.0000	-0.3429	-0.9875	-0.9681	-0.7612	-0.9283	-0.9283	1.0000

COEFFICIENTS OF AUTOCORRELATION

Order	1	2	3	4	5
c	0.9911	0.9779	0.9613	0.9419	0.9202
l	0.8431	0.7062	0.5868	0.4830	0.3928
k	0.9969	0.9887	0.9761	0.9600	0.9410
y	0.9672	0.9341	0.9010	0.8679	0.8352
i	0.8999	0.8104	0.7305	0.6591	0.5951
A	0.9500	0.9025	0.8574	0.8145	0.7738
z	0.9500	0.9025	0.8574	0.8145	0.7738
lambda	0.9911	0.9779	0.9613	0.9419	0.9202



The initial values supplied to Dynare serve as numerical starting points that allow the nonlinear steady state system to converge to a solution. The values were not arbitrary; each initial value corresponds to the analytical steady state relationships derived in parts (i) through (iii). The steady state of the technology process satisfies $z = \bar{z} = 0$ and $A = \bar{A} = e^0 = 1$, because the shock follows $z_t = \rho_A z_{t-1} + \varepsilon_t$ with no long run drift. The steady state labour choice lies between zero and one; here the selected value $l \approx 0.36$ is consistent with the assumption that households devote roughly one third of available time to market work. This value ensures that the marginal rate of substitution between consumption and leisure aligns with the marginal product of labour in steady state. The initial guess for

consumption, $c \approx 0.57$, follows from the resource constraint $\bar{y} = \bar{c} + \bar{i}$ combined with the production function $\bar{y} = \bar{A}\bar{k}^\alpha\bar{l}^{1-\alpha}$ and the depreciation condition $\bar{i} = \delta\bar{k}$. Substituting the calibrated values $\alpha = 0.35$, $\delta = 0.06$, and $\bar{l} \approx 0.36$ yields an implied capital stock near $\bar{k} \approx 2.8$ and output near $\bar{y} \approx 0.74$, which in turn imply a consumption level near $\bar{c} \approx 0.57$. The initial value for investment, $\bar{i} \approx 0.17$, follows from $\delta\bar{k}$ using the same steady state capital stock.

The initial guess for the Lagrange multiplier $\bar{\lambda}$ arises directly from the intratemporal Euler condition, which requires $\lambda = \gamma/c$ in steady state. Using the calibrated $\gamma = 0.4$ and the consumption initial value above yields $\bar{\lambda} \approx 0.7$. These values collectively satisfy the analytical steady state system implied by the planner problem, and supplying them to Dynare ensures that the numerical solver begins sufficiently close to the true steady state so that convergence occurs without instability or cycling.

Dynare delivers the following nonstochastic steady state for the calibrated planner problem: $\bar{c} \approx 0.5727$, $\bar{l} \approx 0.3640$, $\bar{k} \approx 2.8665$, $\bar{y} \approx 0.7447$, $\bar{i} \approx 0.1720$, $\bar{A} = 1$, $\bar{z} = 0$, and $\bar{\lambda} \approx 0.6984$. These values imply that roughly 0.77 of output is consumed and 0.23 is invested in steady state, so that the economy devotes a sizable share of resources to capital accumulation. Labour supply is also close to one third of available time. Moreover, $\bar{\lambda}$ coincides with the marginal utility of consumption, since the first order condition implies $\lambda_t = \gamma/c_t$ and $\gamma/\bar{c} \approx 0.6984$, which confirms that the numerical solution is consistent with the analytical Euler equation derived in parts (i) and (ii). The eigenvalue configuration reported by Dynare shows exactly as many roots outside the unit circle as forward looking variables, so the linearised system has a unique stable equilibrium path in response to a temporary technology disturbance.

A positive innovation raises z_t and therefore $A_t = e^{z_t}$ on impact, which increases the marginal products of capital and labour. The planner responds with a small rise in investment on impact and only a very small increase in labour, after which both variables begin declining as the shock dissipates. Output jumps above steady state on impact and then declines monotonically as the productivity disturbance fades, and consumption also rises on impact and exhibits a small hump shaped adjustment due to consumption smoothing because the Euler condition disciplines intertemporal tradeoffs and induces consumption smoothing

across periods. As the shock dies out geometrically due to $\rho_A < 1$, the marginal products fall back and the economy gradually returns to the original steady state, with capital, as a predetermined state variable, adjusting gradually and generating the delayed hump shaped response observed in the impulse responses.

The theoretical moments reported by Dynare summarise the quantitative properties of the calibrated real business cycle model. The standard deviation of output is 0.0383, whereas the standard deviation of consumption is 0.0258 and the standard deviation of investment is 0.0148. Consumption is therefore smoother than output, consistent with the Euler equation and the intertemporal smoothing motive. Investment displays a lower standard deviation than consumption in this planner allocation because the resource constraint and the absence of adjustment costs restrict the short run reallocation of output between consumption and investment. The standard deviation of capital is 0.1527, which is the largest in absolute terms, but the high mean value of the capital stock implies that capital changes only gradually over time, consistent with the accumulation equation $k_{t+1} = (1 - \delta)k_t + i_t$. The standard deviation of the productivity process and its log representation is 0.0320, reflecting the calibration $\sigma_A = 0.01$ and the persistence parameter $\rho_A = 0.95$. The standard deviation of the Lagrange multiplier is 0.0315, and its magnitude reflects the condition $\lambda_t = \gamma/c_t$, which links fluctuations in marginal utility directly to fluctuations in consumption.

The correlation matrix shows strong procyclical comovement across real variables. The correlation between output and consumption is 0.9681, the correlation between output and investment is 0.8993, and the correlation between output and capital is 0.9165. The correlation between output and labour is 0.5672, indicating that labour increases in expansionary periods when productivity improves. The correlations between output and the productivity variables A and z are both 0.9918, confirming that technology is the principal driver of fluctuations in this economy. The correlation between consumption and the Lagrange multiplier is -1.0000 , exactly as required by the steady state condition $\lambda_t = \gamma/c_t$, which imposes a perfect negative relationship between consumption and marginal utility. The coefficients of autocorrelation confirm the high persistence in all variables generated by the productivity process and the capital accumulation equation. The first order autocorrelation of output is 0.9672, that of consumption is 0.9911, that of investment is 0.8999, and that of capital is

0.9969. The autocorrelation of productivity and its log is 0.9500. Even at order five, the autocorrelations for output, consumption, capital, and productivity remain above 0.7738. These patterns indicate that a temporary technology innovation produces highly persistent but ultimately mean reverting fluctuations in aggregate variables, with consumption adjusting smoothly, investment exhibiting a quick but short lived response, and capital serving as a propagation mechanism.

The impulse response functions confirm the dynamic adjustment of all endogenous variables to a one time unanticipated increase in the technology shock, and each trajectory reflects the structure of the planner's problem and the calibrated parameters. The processes z_t and $A_t = e^{z_t}$ rise immediately on impact and then decline smoothly toward steady state at a geometric rate determined by the persistence parameter $\rho_A = 0.95$, which generates the monotonic decay shown in their respective plots. Output increases on impact and then falls steadily as the productivity disturbance dissipates, producing a strictly decreasing sequence that mirrors the dynamics of A_t through the production function. Consumption rises modestly in the early periods and displays a small hump shaped profile before declining toward steady state, which reflects the smoothing implied by the Euler equation and the planner's preference for a gradual adjustment in intertemporal allocation. Labour decreases on impact and continues along a declining path, indicating that the higher productivity level raises the marginal product of labour sufficiently to allow the planner to reduce hours while maintaining output. Investment rises on impact but then declines monotonically from the first subsequent period onward; the immediate positive jump reflects the higher marginal product of capital, while the rapid decay mirrors the fast adjustment of the optimal investment rate once the shock begins to unwind. Capital exhibits a hump shaped response because it is predetermined on impact and adjusts gradually through the higher initial investment, reaching a peak several periods after the shock before returning to its steady state level. The Lagrange multiplier falls sharply at impact and then moves upward toward steady state, generating a negative hump that is consistent with the relation $\lambda_t = \gamma/c_t$ and the observed rise in consumption during the initial periods. Taken together, the impulse responses illustrate a propagation mechanism in which productivity generates an immediate expansion in economic activity, consumption adjusts smoothly with a modest hump, investment responds

sharply but briefly, labour declines throughout the transition, capital adjusts with delay, and the marginal utility of consumption evolves inversely to consumption over the entire path.

The modification of the capital accumulation equation to

$$k_t = (1 - \delta)k_{t-1} + Z_t i_t$$

introduces an investment specific technology shock that affects the productivity of new investment rather than the current level of total factor productivity in the production function $y_t = A_t k_{t-1}^\alpha l_t^{1-\alpha}$. An increase in Z_t raises the marginal efficiency of investment, because one unit of forgone consumption allocated to investment generates a larger increment in next period capital stock. In terms of the planner problem studied in the earlier parts, the new law of motion implies that the shadow cost of increasing k_t in period t measured in units of current consumption equals λ_t/Z_t rather than λ_t , so a higher value of Z_t reduces the effective price of future capital relative to current consumption and strengthens the incentive to reallocate output toward investment. The Euler condition equating the marginal utility loss from postponing consumption with the discounted marginal utility gain from additional future capital therefore requires a larger investment response when Z_t rises, because the same future return on capital can be obtained with a lower current sacrifice. The immediate effect of a positive investment specific shock is a rise in investment and a fall in contemporaneous consumption, while output responds only through changes in factor inputs, since the production function does not contain Z_t . The prospect of a higher future capital stock increases the expected marginal product of labour in subsequent periods, which raises the expected real wage and encourages additional labour supply. The joint increase in labour input and future capital accumulation generates higher future output and supports higher future consumption, so the economy experiences a persistent expansion driven by the improved efficiency of transforming current resources into productive capital. The mechanism follows the line of research in Greenwood, Hercowitz, and Huffman, where investment specific technological change acts as a disturbance to the marginal efficiency of investment rather than a direct

disturbance to current productivity, and thereby provides a propagation channel that links investment, employment, and output over time within a real business cycle framework.